

519.

ON CURVATURE AND ORTHOGONAL SURFACES.

or conversely

$$\begin{aligned}\xi' &= (1 + d_x \rho \alpha) \xi + d_y \rho \alpha \cdot \eta + d_z \rho \alpha \cdot \zeta, \\ \eta' &= d_x \rho \beta \cdot \xi + (1 + d_y \rho \beta) \eta + d_z \rho \beta \cdot \zeta, \\ \zeta' &= d_x \rho \gamma \cdot \xi + d_y \rho \gamma \cdot \eta + (1 + d_z \rho \gamma) \zeta,\end{aligned}$$

say $\xi', \eta', \zeta' = \xi + \Delta\xi, \eta + \Delta\eta, \zeta + \Delta\zeta$; hence

$$\begin{aligned}X'\xi' + Y'\eta' + Z'\zeta' &= (X - Vd_x\rho)(\xi + \Delta\xi) + \&c. \\ &= X\xi + Y\eta + Z\zeta \\ &\quad + X\Delta\xi + Y\Delta\eta + Z\Delta\zeta - V(\xi d_x\rho + \eta d_y\rho + \zeta d_z\rho),\end{aligned}$$

where second line is

$$\begin{aligned}&(X\alpha + Y\beta + Z\gamma)(\xi d_x\rho + \eta d_y\rho + \zeta d_z\rho) \\ &\rho\{(Xd_x\alpha + Yd_x\beta + Zd_x\gamma)\xi + (Xd_y\alpha + Yd_y\beta + Zd_y\gamma)\eta + (Xd_z\alpha + Yd_z\beta + Zd_z\gamma)\zeta\}.\end{aligned}$$

But

$$\begin{aligned}Xd_x\alpha + Yd_x\beta + Zd_x\gamma &= \frac{\delta X}{V} - \frac{X}{V^3}\delta V = 0, \\ Xd_y\alpha + Yd_y\beta + Zd_y\gamma &= 0, \quad Xd_z\alpha + Yd_z\beta + Zd_z\gamma = 0,\end{aligned}$$

or second line is $= V(\xi d_x\rho + \eta d_y\rho + \zeta d_z\rho)$; and we have therefore

$$X'\xi' + Y'\eta' + Z'\zeta' = X\xi + Y\eta + Z\zeta.$$

We require

$$(A', B', C', F', G', H' \propto \xi', \eta', \zeta')^2;$$

viz., to the first order in ρ , this is

$$= (A', \dots \propto \xi, \eta, \zeta)^2 + 2(A, \dots \propto \Delta\xi, \Delta\eta, \Delta\zeta; \xi, \eta, \zeta).$$

28. Here second line is

$$2\{(A\xi + H\eta + G\zeta)\Delta\xi + (H\xi + B\eta + F\zeta)\Delta\eta + (G\xi + F\eta + C\zeta)\Delta\zeta\};$$

but

$$\begin{aligned}A\xi + H\eta + G\zeta &= Z\delta\eta - Y\delta\zeta + \begin{vmatrix} a & h & g \\ X & Y & Z \\ \xi & \eta & \zeta \end{vmatrix}, \\ H\xi + B\eta + F\zeta &= X\delta\zeta - Z\delta\xi + \begin{vmatrix} h & b & f \\ X & Y & Z \\ \xi & \eta & \zeta \end{vmatrix}, \\ G\xi + F\eta + C\zeta &= Y\delta\xi - X\delta\eta + \begin{vmatrix} g & f & c \\ X & Y & Z \\ \xi & \eta & \zeta \end{vmatrix}.\end{aligned}$$

whence term in $\{ \}$ is

$$\begin{vmatrix} \Delta\xi & \Delta\eta & \Delta\zeta \\ \delta\xi & \delta\eta & \delta\zeta \\ X & Y & Z \end{vmatrix} + \begin{vmatrix} a\Delta\xi + h\Delta\eta + g\Delta\zeta & h\Delta\xi + b\Delta\eta + f\Delta\zeta & g\Delta\xi + f\Delta\eta + c\Delta\zeta \\ X & Y & Z \\ \xi & \eta & \zeta \end{vmatrix},$$

which might be written

$$\begin{vmatrix} \Delta\xi & \Delta\eta & \Delta\zeta \\ \delta\xi & \delta\eta & \delta\zeta \\ X & Y & Z \end{vmatrix} - \begin{vmatrix} \delta\Delta\xi & \delta\Delta\eta & \delta\Delta\zeta \\ \xi & \eta & \zeta \\ X & Y & Z \end{vmatrix},$$

but it is perhaps more convenient to retain the second term in its original form.

29. As regards the first line, we have

$$\begin{aligned} A' &= 2h'Z' - 2g'Y' \\ &= 2(h + \Delta h)(Z - Vd_z\rho) - 2(g + \Delta g)(Y - Vd_y\rho) \\ &= A + 2(Z\Delta h - Y\Delta g) - 2V(hd_z\rho - gd_y\rho), \end{aligned}$$

with similar expressions for the other coefficients. Attending only to the terms of the first order, we thus obtain

$$\begin{aligned} A' &= A + 2(Z\Delta h - Y\Delta g) - 2V(hd_z - gd_y)\rho, \\ B' &= B + 2(X\Delta f - Z\Delta h) - 2V(fd_x - hd_z)\rho, \\ C' &= C + 2(Y\Delta g - X\Delta f) - 2V(gd_y - fd_x)\rho, \\ F' &= F + Y\Delta h - Z\Delta g - X(\Delta b - \Delta c) - V(hd_y - gd_z - (b - c)d_x)\rho, \\ G' &= G + Z\Delta f - X\Delta h - Y(\Delta c - \Delta a) - V(fd_z - hd_x - (c - a)d_y)\rho, \\ H' &= H + X\Delta g - Y\Delta f - Z(\Delta a - \Delta b) - V(gd_x - fd_y - (a - b)d_z)\rho. \end{aligned}$$

Say these are $A' = A + \theta A$, &c., where θ is a functional symbol; we thus have

$$(A', \dots \varrho \xi', \eta', \zeta')^2 = (A, \dots \varrho \xi, \eta, \zeta)^2 + (\theta A, \dots \varrho \xi, \eta, \zeta)^2 + 2(A, \dots \varrho \xi, \eta, \zeta; \Delta\xi, \Delta\eta, \Delta\zeta),$$

which, for shortness, I represent by

$$= (A, \dots \varrho \xi, \eta, \zeta)^2 + (A'', \dots \varrho \xi, \eta, \zeta)^2;$$

and I proceed to complete the calculation of the coefficients $A'', B'', \&c.$

30. We have

$$\begin{aligned} A'' &= \theta A + \text{coeff. } \xi^2 \text{ in } 2[(A\xi + H\eta + G\zeta)\Delta\xi + (H\xi + B\eta + F\zeta)\Delta\eta + (G\xi + F\eta + C\zeta)\Delta\zeta], \\ &= \theta A + 2(Ad_x\rho\alpha + Hd_x\rho\beta + Gd_x\rho\gamma), \end{aligned}$$

that is,

$$A'' = \theta A + \frac{2}{V}(AX + HY + GZ)d_x\rho + 2\rho(Ad_x\alpha + Hd_x\beta + Gd_x\gamma).$$

where coeff. 2ρ is

$$\begin{aligned} &= \frac{Aa + Hh + Gg}{V} - \frac{(AX + HY + GZ)\delta X}{V^3} \\ &= \frac{1}{V} \{ \omega(hZ - gY) + \bar{h}Z - \bar{g}Y \} - \frac{\delta X}{V^3} (Z\delta Y - Y\delta Z). \end{aligned}$$

31. And similarly,

$$\begin{aligned} F'' &= \theta F + (H\alpha + B\beta + F\gamma)d_z\rho + (G\alpha + F\beta + C\gamma)d_y\rho \\ &\quad + \rho \{ (Hd_z\alpha + Bd_z\beta + Fd_z\gamma) + (Gd_y\alpha + Fd_y\beta + Cd_y\gamma) \} \\ &= \theta F + \frac{1}{V} \{ (HX + BY + FZ)d_z\rho + (GX + FY + CZ)d_y\rho \} \\ &\quad + \rho \left\{ \frac{Hg + Bf + Fc}{V} - \frac{(HX + BY + FZ)\delta Z}{V^3} + \frac{Gh + Fb + Cf}{V} \right. \\ &\quad \left. - \frac{(GX + FY + CZ)\delta Y}{V^3} \right\}. \end{aligned}$$

Now

$$Gh + Fb + Cf = \omega(hY - bX) + \bar{h}Y - \bar{b}X + \bar{\omega}X + \bar{a}X + \bar{h}Y + \bar{g}Z,$$

$$Hg + Bf + Fc = -\omega(gZ - cX) - \bar{g}Y + \bar{c}X - \bar{\omega}X - \bar{a}X - \bar{h}Y - \bar{g}Z.$$

Sum is

$$\omega\{hY - gZ - (b - c)X\} + \bar{h}Y - \bar{g}Z - (\bar{b} - \bar{c})X,$$

which is $= \omega F + \bar{h}Y - \bar{g}Z - (\bar{b} - \bar{c})X$; hence

$$F'' = \theta F + (X\delta Z - Z\delta X) \left(\frac{1}{V} d_z\rho - \frac{\rho\delta Z}{V^3} \right) + (Y\delta X - X\delta Y) \left(\frac{1}{V} d_y\rho - \frac{\rho\delta Y}{V^3} \right) + \frac{\rho}{V} \{ \omega F + \bar{F} \}.$$

32. We may write

$$A'' = \theta A + 2 \left(\frac{1}{V} d_x\rho - \frac{\rho\delta X}{V^3} \right) (Z\delta Y - Y\delta Z) + \frac{\rho}{V} \{ \omega A + \bar{A} \},$$

$$B'' = \theta B + 2 \left(\frac{1}{V} d_y\rho - \frac{\rho\delta Y}{V^3} \right) (X\delta Z - Z\delta X) + \frac{\rho}{V} \{ \omega B + \bar{B} \},$$

$$C'' = \theta C + 2 \left(\frac{1}{V} d_z\rho - \frac{\rho\delta Z}{V^3} \right) (Y\delta X - X\delta Y) + \frac{\rho}{V} \{ \omega C + \bar{C} \},$$

$$F'' = \theta F + \left(\frac{1}{V} d_z\rho - \frac{\rho\delta Z}{V^3} \right) (X\delta Z - Z\delta X) + \left(\frac{1}{V} d_y\rho - \frac{\rho\delta Y}{V^3} \right) (Y\delta X - X\delta Y) + \frac{\rho}{V} \{ \omega F + \bar{F} \},$$

$$G'' = \theta G + \left(\frac{1}{V} d_x\rho - \frac{\rho\delta X}{V^3} \right) (Y\delta X - X\delta Y) + \left(\frac{1}{V} d_z\rho - \frac{\rho\delta Z}{V^3} \right) (Z\delta Y - Y\delta Z) + \frac{\rho}{V} \{ \omega G + \bar{G} \},$$

$$H'' = \theta H + \left(\frac{1}{V} d_y\rho - \frac{\rho\delta Y}{V^3} \right) (Z\delta Y - Y\delta Z) + \left(\frac{1}{V} d_x\rho - \frac{\rho\delta X}{V^3} \right) (X\delta Z - Z\delta X) + \frac{\rho}{V} \{ \omega H + \bar{H} \},$$

in which equations $\bar{A}, \bar{B}, \&c.$ are the like functions of $\bar{a}, \bar{b}, \&c.$ that $A, B, \&c.$ are of $a, b, \&c.$; viz. $\bar{A} = 2\bar{h}Z - 2\bar{g}Y, \&c.$

The value of θA is

$$\begin{aligned} \theta A = & 2Z \left(\left\{ -\frac{\rho}{V}(h\omega + \bar{h}) + \frac{\rho}{V^3}\delta X\delta Y \right\} - \frac{1}{V}(\delta Y d_x \rho + \delta X d_y \rho) - V d_x d_y \rho \right) \\ & - 2Y \left(\left\{ -\frac{\rho}{V}(g\omega + \bar{g}) + \frac{\rho}{V^3}\delta Z\delta X \right\} - \frac{1}{V}(\delta Z d_x \rho + \delta X d_z \rho) - V d_x d_z \rho \right) - 2V(hd_z - gd_y)\rho, \end{aligned}$$

which is

$$\begin{aligned} = & -\frac{\rho}{V}(\omega A + \bar{A}) + \frac{2\rho}{V^3}\delta X(Z\delta Y - Y\delta Z) - \frac{2\delta X}{V}(Zd_y - Yd_z)\rho \\ & - 2V(hd_z - gd_y)\rho - \frac{2}{V}(Z\delta Y - Y\delta Z)d_x \rho - 2V(Zd_y - Yd_z)d_x \rho. \end{aligned}$$

Hence the value of A'' is equal to the last-mentioned expression, together with the following terms:

$$+\frac{\rho}{V}(\omega A + \bar{A}) - \frac{2\rho}{V^3}\delta X(Z\delta Y - Y\delta Z) + \frac{2}{V}(Z\delta Y - Y\delta Z)d_x \rho,$$

which destroy certain of the foregoing ones; viz. we have

$$A'' = 2 \left(Vg - \frac{Z\delta X}{V} \right) d_y \rho - 2 \left(Vh - \frac{Y\delta X}{V} \right) d_z \rho - 2V(Zd_y - Yd_z)d_x \rho.$$

33. Similarly, the value of θF is

$$\begin{aligned} \theta F = & Y \left(-\frac{\rho}{V}(h\omega + \bar{h}) + \frac{\rho}{V^3}\delta X\delta Y - \frac{1}{V}(\delta Y d_x \rho + \delta X d_y \rho) - V d_x d_y \rho \right) \\ & - Z \left(-\frac{\rho}{V}(g\omega + \bar{g}) + \frac{\rho}{V^3}\delta Z\delta X - \frac{1}{V}(\delta Z d_x \rho + \delta X d_z \rho) - V d_x d_z \rho \right) \\ & - X \left(-\frac{\rho}{V}\{(b-c)\omega + \bar{b} - \bar{c}\} + \frac{\rho\delta Y^2}{V^3} - \frac{\rho\delta Z^2}{V^3} - \frac{2}{V}\delta Y d_y \rho + \frac{2}{V}\delta Z d_z \rho - V d_y^2 \rho + V d_z^2 \rho \right) \\ & - V(hd_y - gd_z - (b-c)d_x)\rho, \end{aligned}$$

which is

$$\begin{aligned} = & \frac{\rho}{V}(-F\omega - \bar{F}) + \frac{\rho\delta Z}{V^3}(X\delta Z - Z\delta X) + \frac{\rho\delta Y}{V^3}(Y\delta X - X\delta Y) \\ & + \left\{ -\frac{1}{V}(Y\delta Y - Z\delta Z) + V(b-c) \right\} d_x \rho \\ & + \left\{ -\frac{Y}{V}\delta X + \frac{2X}{V}\delta Y - Vh \right\} d_y \rho \\ & + \left\{ \frac{Z}{V}\delta X + \frac{2X}{V}\delta Y + Vg \right\} d_z \rho \\ & + (-VY d_x d_y + VZ d_x d_z + VX d_y^2 - VX d_z^2)\rho. \end{aligned}$$

Hence F'' is equal to the foregoing expression, together with the following terms:

$$\begin{aligned}
 & + \frac{\rho}{V}(F\omega + \bar{F}) - \frac{\rho\delta Z}{V^3}(X\delta Z - Z\delta X) - \frac{\rho\delta Y}{V^3}(Y\delta X - X\delta Y) \\
 & + \frac{1}{V}(Y\delta X - X\delta Y)d_y\rho + \frac{1}{V}(X\delta Z - Z\delta X)d_z\rho,
 \end{aligned}$$

which destroy certain of the foregoing terms; viz. we thus have

$$\begin{aligned}
 F'' = & \left\{ -\frac{1}{V}(Y\delta Y - Z\delta Z) + V(b - c) \right\} d_x\rho + \left\{ \frac{X}{V}\delta Y - Vh \right\} d_y\rho + \left\{ -\frac{X}{V}\delta Z + Vg \right\} d_z\rho \\
 & + V(-Yd_xd_y + Zd_xd_z + Xd_y^2 - Xd_z^2)\rho.
 \end{aligned}$$

34. We thus have

$$\begin{aligned}
 A'' = & 2\left(Vg - \frac{Z\delta X}{V}\right)d_y\rho - 2\left(Vh - \frac{Y\delta X}{V}\right)d_z\rho + 2V(-Zd_yd_x + Yd_zd_x)\rho, \\
 B'' = & -2\left(Vf - \frac{Z\delta Y}{V}\right)d_x\rho + 2\left(Vh - \frac{X\delta Y}{V}\right)d_z\rho + 2V(-Xd_zd_y + Zd_xd_y)\rho, \\
 C'' = & 2\left(Vf - \frac{Y\delta Z}{V}\right)d_x\rho - 2\left(Vg - \frac{X\delta Z}{V}\right)d_y\rho + 2V(-Yd_xd_z + Xd_yd_z)\rho, \\
 F'' = & \left\{ V(b - c) - \frac{1}{V}(Y\delta Y - Z\delta Z) \right\} d_x\rho - \left(Vh - \frac{X\delta Y}{V} \right) d_y\rho + \left(Vg - \frac{X\delta Z}{V} \right) d_z\rho \\
 & + V(-Yd_xd_y + Zd_xd_z + Xd_y^2 - Xd_z^2)\rho, \\
 G'' = & \left(Vh - \frac{Y\delta X}{V} \right) d_x\rho + \left\{ V(c - a) - \frac{1}{V}(Z\delta Z - X\delta X) \right\} d_y\rho - \left(Vf - \frac{Y\delta Z}{V} \right) d_z\rho \\
 & + V(-Zd_yd_z + Xd_yd_x + Yd_z^2 - Yd_x^2)\rho, \\
 H'' = & -\left(Vg - \frac{Z\delta X}{V} \right) d_x\rho - \left(Vf - \frac{Z\delta Y}{V} \right) d_y\rho + \left\{ V(a - b) - \frac{1}{V}(X\delta X - Y\delta Y) \right\} d_z\rho \\
 & + V(-Xd_zd_x + Yd_zd_y + Zd_x^2 - Zd_y^2)\rho.
 \end{aligned}$$

35. It will be recollected that we have

$$X'\xi' + Y'\eta' + Z'\zeta' = X\xi + Y\eta + Z\zeta;$$

by what precedes it appears that for the given surface the principal tangents are determined by the equations

$$(A, \dots \varrho \xi, \eta, \zeta)^2 = 0, \quad X\xi + Y\eta + Z\zeta = 0,$$

and that the lines which (in the tangent plane of the given surface) correspond to the principal tangents of the corresponding point of the vicinal surface are determined by the equations

$$(A, \dots \varrho \xi, \eta, \zeta)^2 + (A'', \dots \varrho \xi, \eta, \zeta)^2 = 0, \quad X\xi + Y\eta + Z\zeta = 0.$$

Condition that the two surfaces may belong to an Orthogonal System.

Art. Nos. 36 to 41.

36. The condition in order that the two surfaces may belong to an orthogonal system is that the principal tangents shall correspond, or, what is the same thing, the lines which (in the tangent plane of the given surface) correspond to the principal tangents of the vicinal surface must be the principal tangents of the given surface. When this is the case, the plane and cone $X\xi + Y\eta + Z\zeta = 0$, $(A'', \dots \oslash \xi, \eta, \zeta)^2 = 0$ intersect in the principal tangents, and this is therefore the required condition.

The plane $X\xi + Y\eta + Z\zeta = 0$ meets the cone $(A'', \dots \oslash \xi, \eta, \zeta)^2 = 0$ in the principal tangents, that is, in a pair of lines harmonically related to the circular lines and also to the chief tangents. Forming then the coefficients $(A''), (B''), (C''), (F''), (G''), (H'')$ from $A'', \&c.$ in the same way as $(A)\&c.$ are formed from $A, \&c.$, that is, writing $(A'') = B''Z^2 + C''Y^2 - 2F''YZ, \&c.$, the conditions are

$$(A'') + (B'') + (C'') = 0, \quad ((A''), \dots \oslash a, \dots) = 0,$$

or, what is the same thing,

$$(A'', \dots \oslash (a), \dots) = 0.$$

The former of these, as about to be shown, is satisfied identically; we have therefore the second of them, say $(A'', \dots \oslash (a), \dots) = 0$, as the required condition.

37. We have

$$(A'') + (B'') + (C'') = (A'' + B'' + C'')V^2 - (A'', \dots \oslash X, Y, Z)^2,$$

$$A'' + B'' + C'' = \frac{2}{V} \{ (Z\delta Y - Y\delta Z)d_x\rho + (X\delta Z - Z\delta X)d_y\rho + (Y\delta X - X\delta Y)d_z\rho \}.$$

Forming next the expressions of $A''X + H''Y + G''Z$, &c., and, for convenience, writing down separately the terms which involve the second differential coefficients of ρ , we have

$$\begin{aligned} A''X + H''Y + G''Z &= d_x\rho \cdot V(hZ - gY) + d_y\rho[V\delta Z - Z\delta V + V(gX - aZ)] \\ &\quad + d_z\rho[-(V\delta Y - Y\delta V) - V(hX - aY)], \end{aligned}$$

$$\begin{aligned} H''X + B''Y + F''Z &= d_x\rho[-(V\delta Z - Z\delta V) - V(fY - bZ)] \\ &\quad + d_y\rho \cdot V(fX - hZ) + d_z\rho[(V\delta X - X\delta V) + V(hY - bX)], \end{aligned}$$

$$\begin{aligned} G''X + F''Y + C''Z &= d_x\rho[V\delta Y - Y\delta V + V(fZ - cY)] \\ &\quad + d_y\rho[-(V\delta X - X\delta V) - V(gZ - cX)] + d_z\rho \cdot V(gY - fX), \end{aligned}$$

where δV stands for $\frac{1}{V}(X\delta X + Y\delta Y + Z\delta Z)$, and where the three expressions contain also the following terms respectively:

$$\{-YZd_y^2 + YZd_z^2 + (Y^2 - Z^2)d_yd_z + XYd_zd_x - XZd_xd_y\}\rho,$$

$$\{ZXd_x^2 - ZXd_z^2 - XYd_yd_z + (Z^2 - X^2)d_zd_x + YZd_xd_y\}\rho,$$

$$\{-XYd_x^2 + XYd_y^2 + XZd_yd_z - YZd_zd_x + (X^2 - Y^2)d_xd_y\}\rho.$$

Multiplying by X, Y, Z , and adding, the terms which contain the second differential coefficients disappear, and we obtain

$$(A'', \dots \varrho X, Y, Z)^2 = 2V\{(Z\delta Y - Y\delta Z)d_x\rho + (X\delta Z - Z\delta X)d_y\rho + (Y\delta X - X\delta Y)d_z\rho\};$$

so that, attending to the above value of $A'' + B'' + C''$, we have the required equation

$$(A'') + (B'') + (C'') = 0.$$

38. Proceeding now to form the value of $(A'', \dots \varrho (a), \dots)$, that is,

$$A''(a) + B''(b) + C''(c) + 2F''(f) + 2G''(g) + 2H''(h),$$

it will be shown that the terms involving the first differential coefficients of ρ vanish of themselves; as regards those containing the second differential coefficients, forming the auxiliary equations

$$\begin{aligned} (A) &= 2(h)Z - 2(g)Y, \\ (B) &= 2(f)X - 2(h)Z, \\ (C) &= 2(g)Y - 2(f)X, \\ (F) &= (h)Y - (g)Z - ((b) - (c))X, \\ (G) &= (f)Z - (h)X - ((c) - (a))Y, \\ (H) &= (g)X - (f)Y - ((a) - (b))Z, \end{aligned}$$

we find without difficulty that the terms in question (being, in fact, the complete value of the expression) are

$$= V((A), \dots \varrho d_x, d_y, d_z)^2 \rho.$$

39. As regards the terms involving the first differential coefficients, observe that the whole coefficient of $d_x\rho$ is

$$\begin{aligned} -2(b) \left(Vf - \frac{Z\delta Y}{V} \right) + 2(c) \left(Vg - \frac{Y\delta Z}{V} \right) + 2(f) \left(V(b - c) - \frac{1}{V}(Y\delta Y - Z\delta Z) \right) \\ + 2(g) \left(Vh - \frac{Y\delta X}{V} \right) - 2(h) \left(Vg - \frac{Z\delta X}{V} \right), \end{aligned}$$

which is

$$\begin{aligned} &= 2V\{(g)h + (f)b + (c)g - ((h)g + (b)f + (f)c)\} \\ &+ \frac{2}{V}\{Z((h)\delta X + (b)\delta Y + (f)\delta Z) - Y((g)\delta X + (f)\delta Y + (c)\delta Z)\}. \end{aligned}$$

40. The reduction depends on the following auxiliary formulae:

$$\begin{aligned} a(a) + h(h) + g(g) &= V\bar{\delta}V - X\bar{\delta}X, & a(h) + h(b) + g(f) &= -X\bar{\delta}Y, \\ a(g) + h(f) + g(c) &= -X\bar{\delta}Z, & h(a) + b(h) + f(g) &= -Y\bar{\delta}X, \\ h(h) + b(b) + f(f) &= V\bar{\delta}V - Y\bar{\delta}Y, & h(g) + b(f) + f(c) &= -Y\bar{\delta}Z, \\ g(a) + f(h) + c(g) &= -Z\bar{\delta}X, & g(h) + f(b) + c(f) &= -Z\bar{\delta}Y, \\ g(g) + f(f) + c(c) &= V\bar{\delta}V - Z\bar{\delta}Z. \end{aligned}$$

where, for shortness, I have written $\bar{\delta}X, \bar{\delta}Y, \bar{\delta}Z$ to stand for $\bar{a}X + \bar{h}Y + \bar{g}Z$, $\bar{h}X + \bar{b}Y + \bar{f}Z$, $\bar{g}X + \bar{f}Y + \bar{c}Z$ respectively, and $V\bar{\delta}V$ for

$$X\bar{\delta}X + Y\bar{\delta}Y + Z\bar{\delta}Z, \quad (= (\bar{a}, \dots \varrho X, Y, Z)^2).$$

From these we immediately have

$$\begin{aligned} (a)\delta X + (h)\delta Y + (g)\delta Z &= V(X\bar{\delta}V - V\bar{\delta}X), \\ (h)\delta X + (b)\delta Y + (f)\delta Z &= V(Y\bar{\delta}V - V\bar{\delta}Y), \\ (g)\delta X + (f)\delta Y + (c)\delta Z &= V(Z\bar{\delta}V - V\bar{\delta}Z). \end{aligned}$$

Hence, in the coefficient of $d_x\rho$, the first line is

$$= 2V(-Y\bar{\delta}Z + Z\bar{\delta}Y),$$

and the second line is

$$= \frac{2}{V}\{VZ(Y\bar{\delta}V - V\bar{\delta}Y) - VY(Z\bar{\delta}V - V\bar{\delta}Z)\} = 2V(Y\bar{\delta}Z - Z\bar{\delta}Y);$$

so that the sum, or whole coefficient of $d_x\rho$, is $= 0$. Similarly, the coefficients of $d_y\rho$ and $d_z\rho$ are each $= 0$.

41. We have thus arrived at the equation

$$((A), \dots \varrho d_x, d_y, d_z)^2 \rho = 0$$

as the condition to be satisfied by the normal distance ρ in order that the given surface and the vicinal surface may belong to an orthogonal system, viz. this is a partial differential equation of the second order, its coefficients being given functions of $X, Y, Z, a, b, c, f, g, h$, the first and second differential coefficients of r (where $r = r(x, y, z)$ is the equation of the given surface).

The equation, it is clear, may also be written in the two forms

$$(A, \dots \varrho Zd_y - Yd_z, Xd_z - Zd_x, Yd_x - Xd_y)^2 \rho = 0,$$

and

$$\begin{vmatrix} P & Q & R \\ aP + hQ + gR & hP + bQ + fR & gP + fQ + cR \\ X & Y & Z \end{vmatrix} \rho = 0,$$

if, for shortness, P, Q, R are written to denote $Zd_y - Yd_z$, $Xd_z - Zd_x$, $Yd_x - Xd_y$ respectively, it being understood that in each of these forms the d_x, d_y, d_z operate on the ρ only.

Condition that a family of surfaces may belong to an Orthogonal System.

Art. Nos. 42 to 49.

42. We pass at once to the condition in order that the family of surfaces

$$r - r(x, y, z) = 0$$

may belong to an orthogonal system, viz. when the vicinal surface belongs to the family, we have

$$\rho \text{ proportional to } \frac{1}{V} \left(= \frac{1}{\sqrt{X^2 + Y^2 + Z^2}} \right),$$

and the condition is

$$((A), \dots \oslash d_x, d_y, d_z)^2 \frac{1}{V} = 0,$$

where r is a function of (x, y, z) , the first and second differential coefficients of which are $X, Y, Z, a, b, c, f, g, h$; and the equation is thus a partial differential equation of the third order satisfied by r . The form is by no means an inconvenient one, but it admits of further reduction.

43. We have $d_x \frac{1}{V}, d_y \frac{1}{V}, d_z \frac{1}{V}$ equal to $-\frac{1}{V^3} \delta X, -\frac{1}{V^3} \delta Y, -\frac{1}{V^3} \delta Z$ respectively, and thence

$$d_x^2 \frac{1}{V} = -\frac{1}{V^3} (a^2 + h^2 + g^2 + \delta a) + \frac{3}{V^5} (\delta X)^2,$$

$$d_y d_z \frac{1}{V} = -\frac{1}{V^3} (gh + bf + cf + \delta f) + \frac{3}{V^5} \delta Y \delta Z,$$

or, as these may be written,

$$d_x^2 \frac{1}{V} = -\frac{1}{V^3} (a\omega - \bar{\omega} + \bar{a} + \delta a) + \frac{3}{V^5} (\delta X)^2,$$

$$d_y d_z \frac{1}{V} = -\frac{1}{V^3} (f\omega + \bar{f} + \delta f) + \frac{3}{V^5} \delta Y \delta Z,$$

with the like values for $d_y^2 \frac{1}{V}$, &c. Substituting, the equation contains a term multiplied by ω , viz. this is

$$-\frac{1}{V^3} \omega ((A), \dots \oslash a, \dots),$$

which vanishes; and a term multiplied by $\bar{\omega}$, viz. this is

$$\frac{1}{V^3} \bar{\omega} \{(A) + (B) + (C)\},$$

which also vanishes. Writing down the remaining terms, and multiplying the whole by $-V^3$, the equation becomes

$$((A), \dots \oslash \bar{a}, \dots) + ((A), \dots \oslash \delta a, \dots) - \frac{3}{V^2} ((A), \dots \oslash \delta X, \delta Y, \delta Z)^2 = 0.$$

44. The last term admits of reduction; from the equations

$$(A) = -AV^2 + 2XZ\delta Y - 2XY\delta Z, \quad \&c.,$$

we find

$$(A)\delta X + (H)\delta Y + (G)\delta Z = -V^2(A\delta X + H\delta Y + G\delta Z) + V\delta V(Z\delta Y - Y\delta Z),$$

$$(H)\delta X + (B)\delta Y + (F)\delta Z = -V^2(H\delta X + B\delta Y + F\delta Z) + V\delta V(X\delta Z - Z\delta X),$$

$$(G)\delta X + (F)\delta Y + (C)\delta Z = -V^2(G\delta X + F\delta Y + C\delta Z) + V\delta V(Y\delta X - X\delta Y),$$

and hence

$$((A), \dots \varrho \delta X, \delta Y, \delta Z)^2 = -V^2 (A, \dots \varrho \delta X, \delta Y, \delta Z)^2;$$

wherefore the equation becomes

$$((A), \dots \varrho \bar{a}, \dots) + ((A), \dots \varrho \delta a, \dots) + 3(A, \dots \varrho \delta X, \delta Y, \delta Z)^2 = 0.$$

45. It will be shown that we have identically

$$((A), \dots \varrho \bar{a}, \dots) = -(A, \dots \varrho \delta X, \delta Y, \delta Z)^2 = 2 \begin{vmatrix} \delta X & \delta Y & \delta Z \\ X & Y & Z \\ \bar{\delta} X & \bar{\delta} Y & \bar{\delta} Z \end{vmatrix}.$$

The partial differential equation thus assumes the form

$$((A), \dots \varrho \delta a, \dots) + \Omega = 0,$$

where Ω may be expressed indifferently in the three forms,

$$= +2(A, \dots \varrho \bar{a}, \dots), \quad = +2(A, \dots \varrho \delta X, \delta Y, \delta Z)^2,$$

$$= -4 \begin{vmatrix} \delta X & \delta Y & \delta Z \\ X & Y & Z \\ \bar{\delta} X & \bar{\delta} Y & \bar{\delta} Z \end{vmatrix}.$$

46. Taking the first of these, the partial differential equation is

$$((A), \dots \varrho \delta a, \dots) - 2((A), \dots \varrho \bar{a}, \dots) = 0;$$

or, written at full length, it is

$$(A)\delta a + (B)\delta b + (C)\delta c + 2(F)\delta f + 2(G)\delta g + 2(H)\delta h \\ - 2\{(A)\bar{a} + (B)\bar{b} + (C)\bar{c} + 2(F)\bar{f} + 2(G)\bar{g} + 2(H)\bar{h}\} = 0,$$

where the coefficients are given functions of $X, Y, Z, a, b, c, f, g, h$, the first and second differential coefficients of r ; and δ is written to denote $Xd_x + Yd_y + Zd_z$.

47. It remains to prove the above-mentioned identities.

To reduce the term $(A, \dots \oslash \delta X, \delta Y, \delta Z)^2$, we have

$$\begin{aligned}
 & A\delta X + H\delta Y + G\delta Z \\
 &= A(aX + hY + gZ) + H(hX + bY + fZ) + G(gX + fY + cZ) \\
 &= X\{\omega(hZ - gY) + \bar{h}Z - \bar{g}Y\} + Y\{-\omega(fY - bZ) - (\bar{f}Y - \bar{b}Z) - \bar{\omega}Z - \delta Z\} \\
 &\quad + Z\{\omega(fZ - cY) + fZ - \bar{c}Y + \bar{\omega}Y + \delta Z\} \\
 &= \omega(Z\delta Y - Y\delta Z) + (Z\bar{\delta}Y - Y\bar{\delta}Z) + (Z\delta Y - Y\delta Z),
 \end{aligned}$$

that is,

$$A\delta X + H\delta Y + G\delta Z = \omega(Z\delta Y - Y\delta Z) + 2(Z\bar{\delta}Y - Y\bar{\delta}Z);$$

and similarly

$$H\delta X + B\delta Y + F\delta Z = \omega(X\delta Z - Z\delta X) + 2(X\bar{\delta}Z - Z\bar{\delta}X),$$

$$G\delta X + F\delta Y + C\delta Z = \omega(Y\delta X - X\delta Y) + 2(Y\bar{\delta}X - X\bar{\delta}Y),$$

whence

$$(A, \dots \oslash \delta X, \delta Y, \delta Z)^2 = -2 \begin{vmatrix} \delta X & \delta Y & \delta Z \\ X & Y & Z \\ \bar{\delta} X & \bar{\delta} Y & \bar{\delta} Z \end{vmatrix}.$$

48. Now, from the equations $AX + HY + GZ = Z\delta Y - Y\delta Z$, &c. we have for the value of twice the foregoing determinant

$$\begin{aligned}
 2 \det. &= 2\{(\bar{a}X + \bar{h}Y + \bar{g}Z)(AX + HY + GZ) \\
 &\quad + (\bar{h}X + \bar{b}Y + \bar{f}Z)(HX + BY + FZ) + (\bar{g}X + \bar{f}Y + \bar{c}Z)(GX + FY + CZ)\};
 \end{aligned}$$

and subtracting herefrom the function $((A), \dots \oslash \bar{a}, \dots)$, which is

$$\begin{aligned}
 &= (BZ^2 + CY^2 - 2FYZ)\bar{a} + (CX^2 + AZ^2 - 2GZX)\bar{b} + (AY^2 + BX^2 - 2HYZ)\bar{c} \\
 &\quad + 2(-AYZ - FX^2 + GXY + HXZ)\bar{f} + 2(-BZX + FXY - GY^2 + HYZ)\bar{g} \\
 &\quad + 2(-CXY + FXZ + GYZ - HZ^2)\bar{h},
 \end{aligned}$$

the difference is found to be

$$\begin{aligned}
 &= \bar{a}\{(A, \dots \oslash X, Y, Z)^2 + AV^2\} + \bar{b}\{(A, \dots \oslash X, Y, Z)^2 + BV^2\} + \bar{c}\{(A, \dots \oslash X, Y, Z)^2 + CV^2\} \\
 &\quad + 2\bar{f}\{(A + B + C)YZ + FV^2\} + 2\bar{g}\{(A + B + C)ZX + GV^2\} + 2\bar{h}\{(A + B + C)XY + HV^2\},
 \end{aligned}$$

which, on account of $(A, \dots \mathfrak{Q} X, Y, Z)^2 = 0$, and $A + B + C = 0$, reduces itself to

$$(A, \dots \mathfrak{Q} \bar{a}, \dots) V^2.$$

49. We have

$$\begin{aligned} A\bar{a} + H\bar{h} + G\bar{g} &= \bar{a}(2hZ - 2gY) \\ &\quad + \bar{h}(gX - fY - (a - b)Z) \\ &\quad + \bar{g}(fZ - hX - (c - a)Y) \\ &= X(\bar{h}g - \bar{g}h) \\ &\quad + Y(\bar{g}a - \bar{a}g - (g\bar{a} + f\bar{h} + c\bar{g})) \\ &\quad + Z(\bar{h}a - \bar{a}h + (h\bar{a} + b\bar{h} + f\bar{g})); \end{aligned}$$

or, observing that in the coefficients of Y and Z the second terms each vanish, this is

$$A\bar{a} + H\bar{h} + G\bar{g} = X(\bar{h}g - \bar{g}h) + Y(\bar{g}a - \bar{a}g) + Z(\bar{a}h - \bar{h}a);$$

and similarly

$$\begin{aligned} H\bar{h} + B\bar{b} + F\bar{f} &= X(\bar{b}f - \bar{f}b) + Y(\bar{f}h - \bar{h}f) + Z(\bar{h}b - \bar{b}h), \\ G\bar{g} + F\bar{f} + C\bar{c} &= X(\bar{f}c - \bar{c}f) + Y(\bar{c}g - \bar{g}c) + Z(\bar{g}f - \bar{f}g). \end{aligned}$$

Adding these equations, the coefficient of X is the difference of two expressions each of which vanishes; and the like as regards the coefficients of Y and Z ; that is, we have

$$(A, \dots \mathfrak{Q} \bar{a}, \dots) = 0;$$

and consequently

$$2 \begin{vmatrix} \delta X & \delta Y & \delta Z \\ X & Y & Z \\ \bar{\delta} X & \bar{\delta} Y & \bar{\delta} Z \end{vmatrix} = ((A), \dots \mathfrak{Q} \bar{a}, \dots) = -(A, \dots \mathfrak{Q} \delta X, \delta Y, \delta Z)^2,$$

the required relation.

ON THE CENTRO-SURFACE OF AN ELLIPSOID.

[From the *Transactions of the Cambridge Philosophical Society*, vol. XII. Part I. (1873), pp. 319–365. Read March 7, 1870.]

THE Centro-surface of any given surface is the locus of the centres of curvature of the given surface, or say it is the locus of the intersections of consecutive normals, (the normals which intersect the normal at any particular point of the surface being those at the consecutive points along the two curves of curvature respectively which pass through the point on the surface). The terms, normal, centre of curvature, curve of curvature, may be understood in their ordinary sense, or in the generalised sense referring to the case where the Absolute (instead of being the imaginary circle at infinity) is any quadric surface whatever; viz. the normal at any point of a surface is here the line joining that point with the pole of the tangent plane in respect of the quadric surface called the Absolute: and of course the centre of curvature and curve of curvature refer to the normal as just defined.

The question of the centro-surface of a quadric surface has been considered in the two points of view, viz. 1°, when the terms “normal,” &c. are used in the ordinary sense, and the equation of the quadric surface (assumed to be an ellipsoid) is taken to be

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1;$$

2°, when the Absolute is the surface $X^2 + Y^2 + Z^2 + W^2 = 0$, and the equation of the quadric surface is taken to be

$$\alpha X^2 + \beta Y^2 + \gamma Z^2 + \delta W^2 = 0 :$$

in the first of them by Salmon, *Quart. Math. Jour.* t. II. pp. 217–222 (1858), and in the second by Clebsch, *Crelle*, t. LXII. pp. 64–107 (1863): see also Salmon’s *Solid Geometry*, 2nd Ed. 1865, pp. 143, 402, &c. In the present Memoir, as shown by the title, the quadric surface is taken to be an Ellipsoid; and the question is considered exclusively from the first point of view: the theory is further developed in various respects, and in particular as regards the nodal curve upon the centro-surface: the distinction of real and